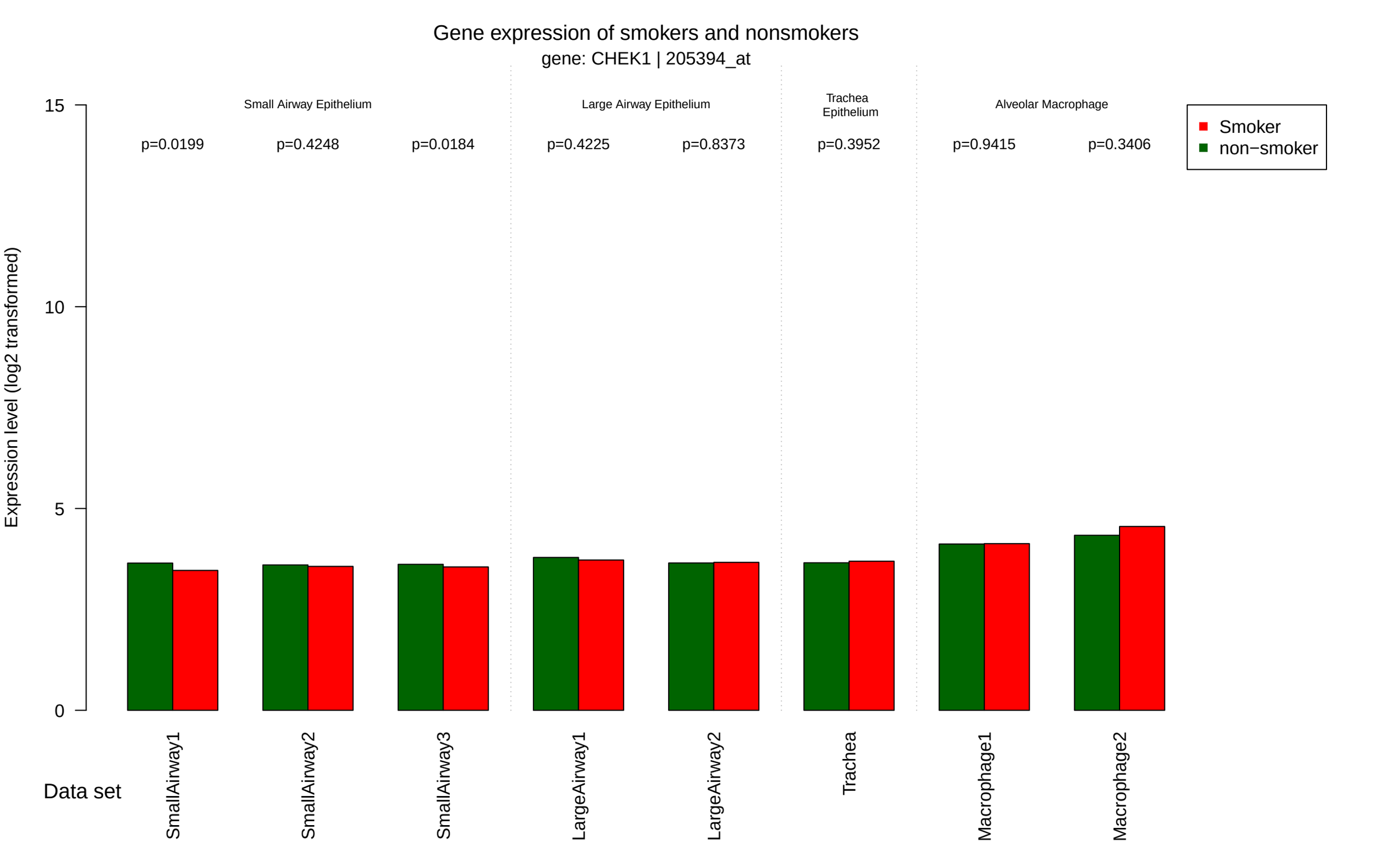
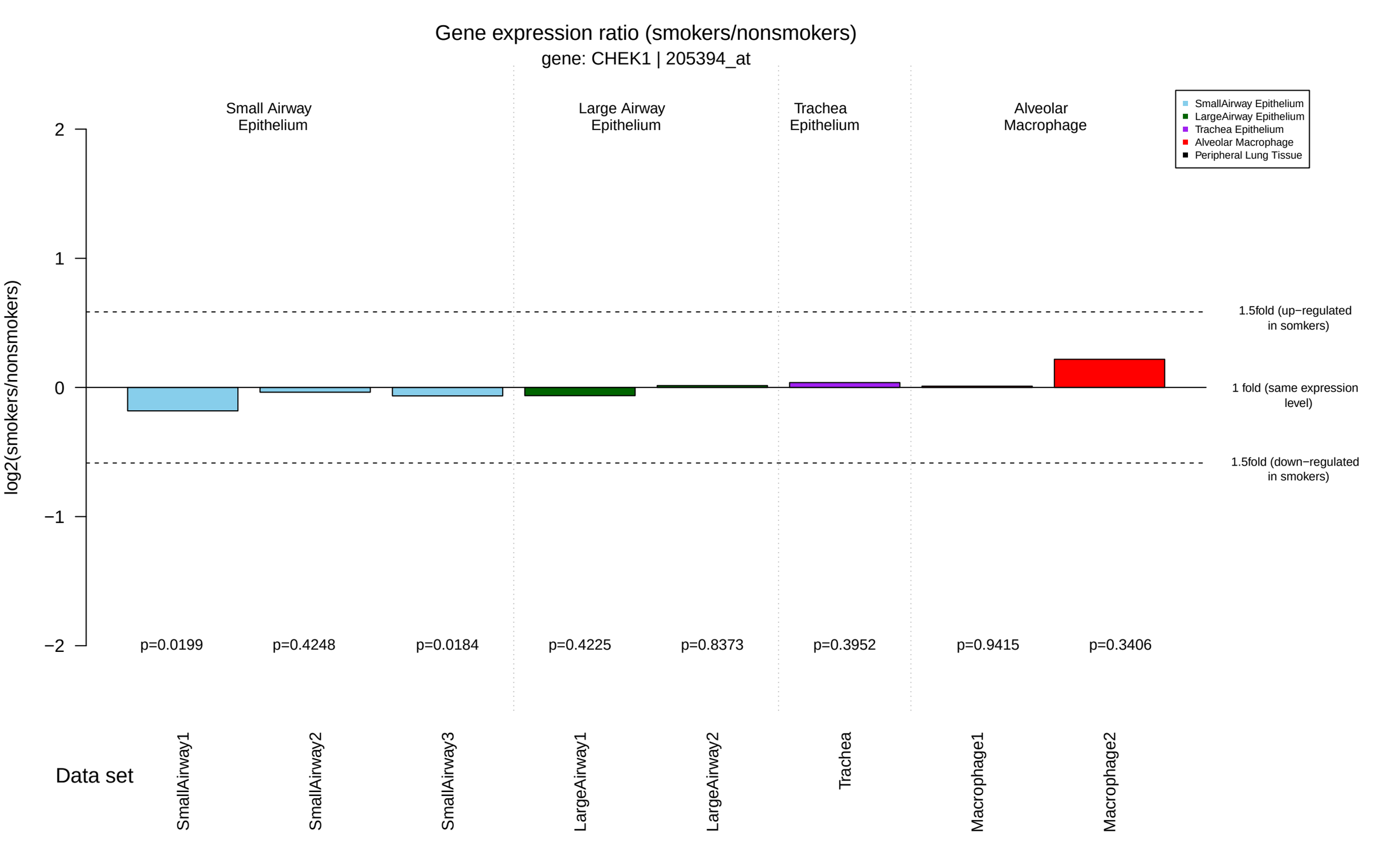
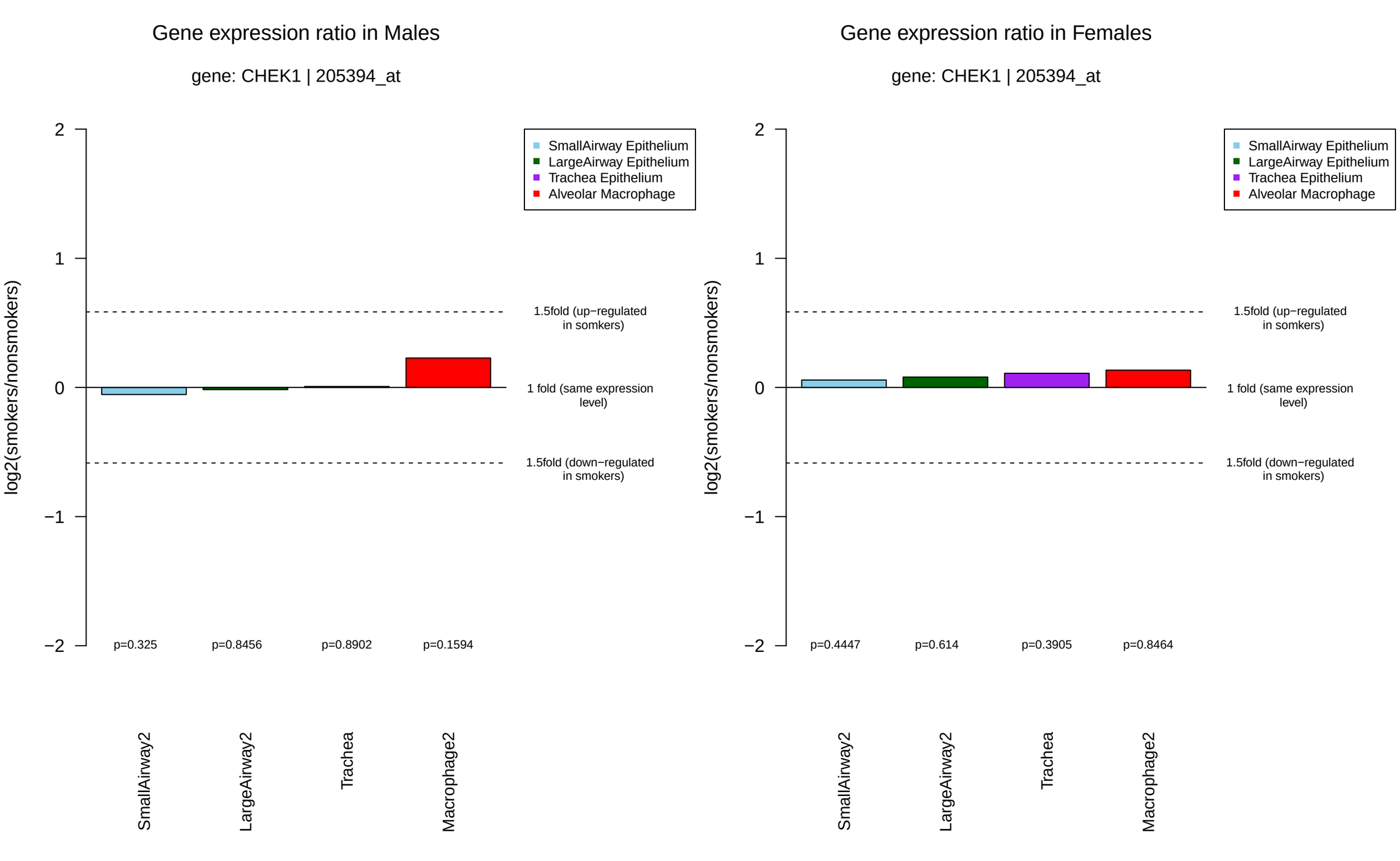
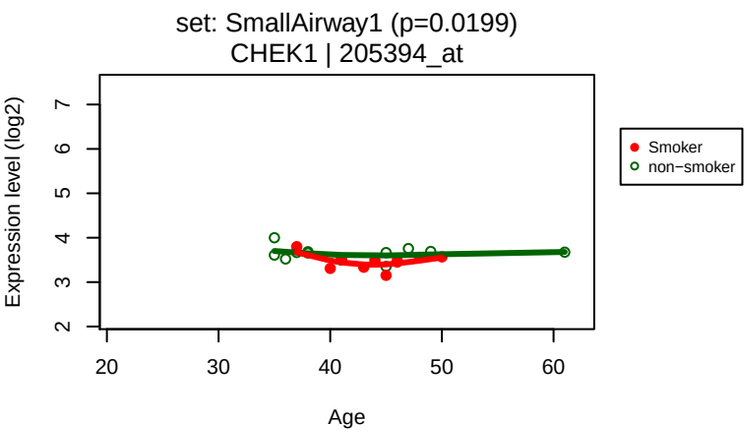
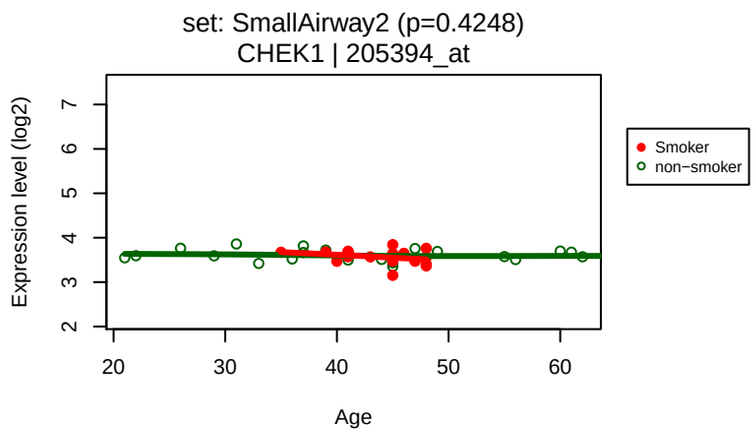
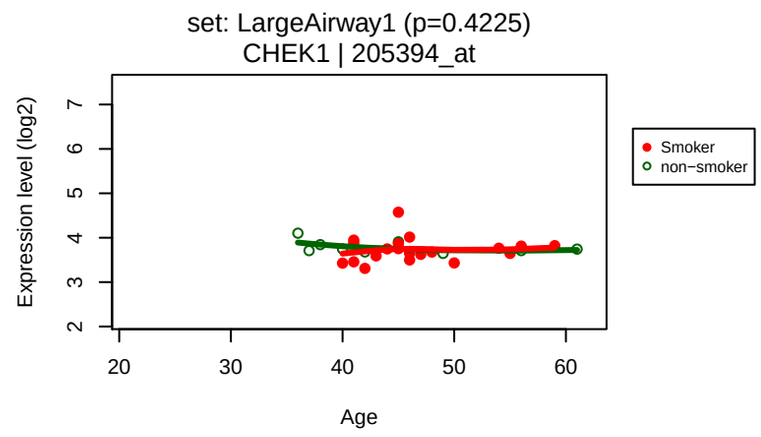

Gene expression of smokers and nonsmokers
gene: CHEK1 | 205394_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0199
p=0.4248
p=0.0184
p=0.4225
p=0.8373
p=0.3952
p=0.9415
p=0.3406
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

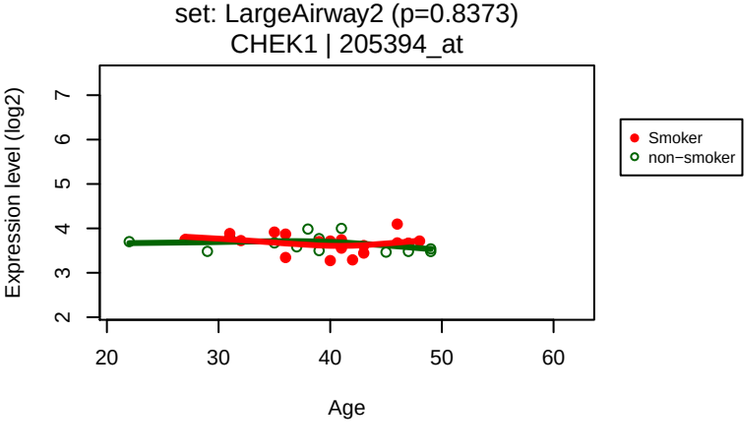

Gene expression ratio (smokers/nonsmokers)
gene: CHEK1 | 205394_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0199
p=0.4248
p=0.0184
p=0.4225
p=0.8373
p=0.3952
p=0.9415
p=0.3406
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

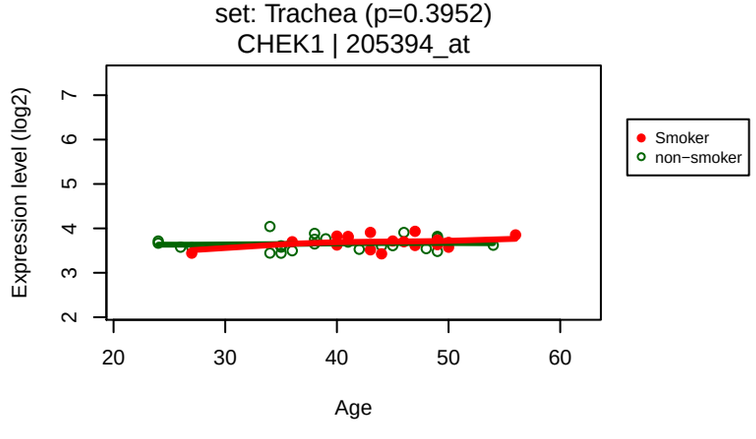

Gene expression ratio in Males
gene: CHEK1 | 205394_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.325
p=0.8456
p=0.8902
p=0.1594
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CHEK1 | 205394_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4447
p=0.614
p=0.3905
p=0.8464
SmallAirway2
LargeAirway2
Trachea
Macrophage2

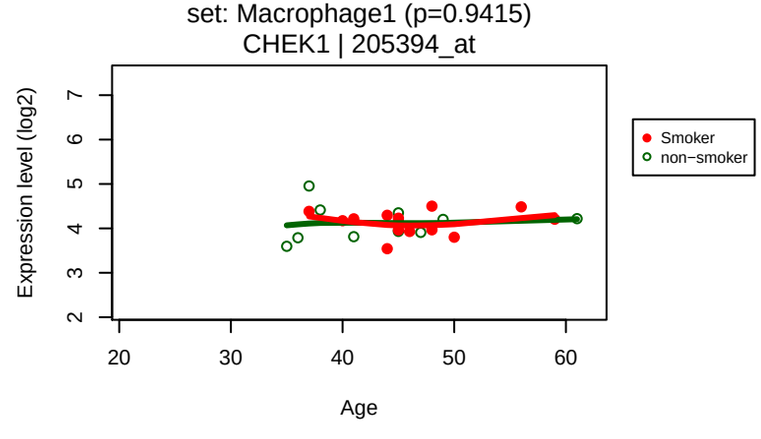

set: SmallAirway1 (p=0.0199)
CHEK1 | 205394_at
Expression level (log2)
Age
Smoker
non−smoker

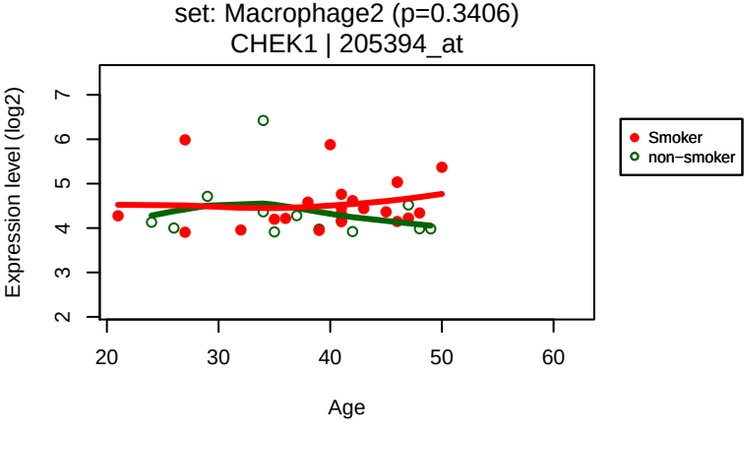

set: SmallAirway2 (p=0.4248)
CHEK1 | 205394_at
Expression level (log2)
Age
Smoker
non−smoker

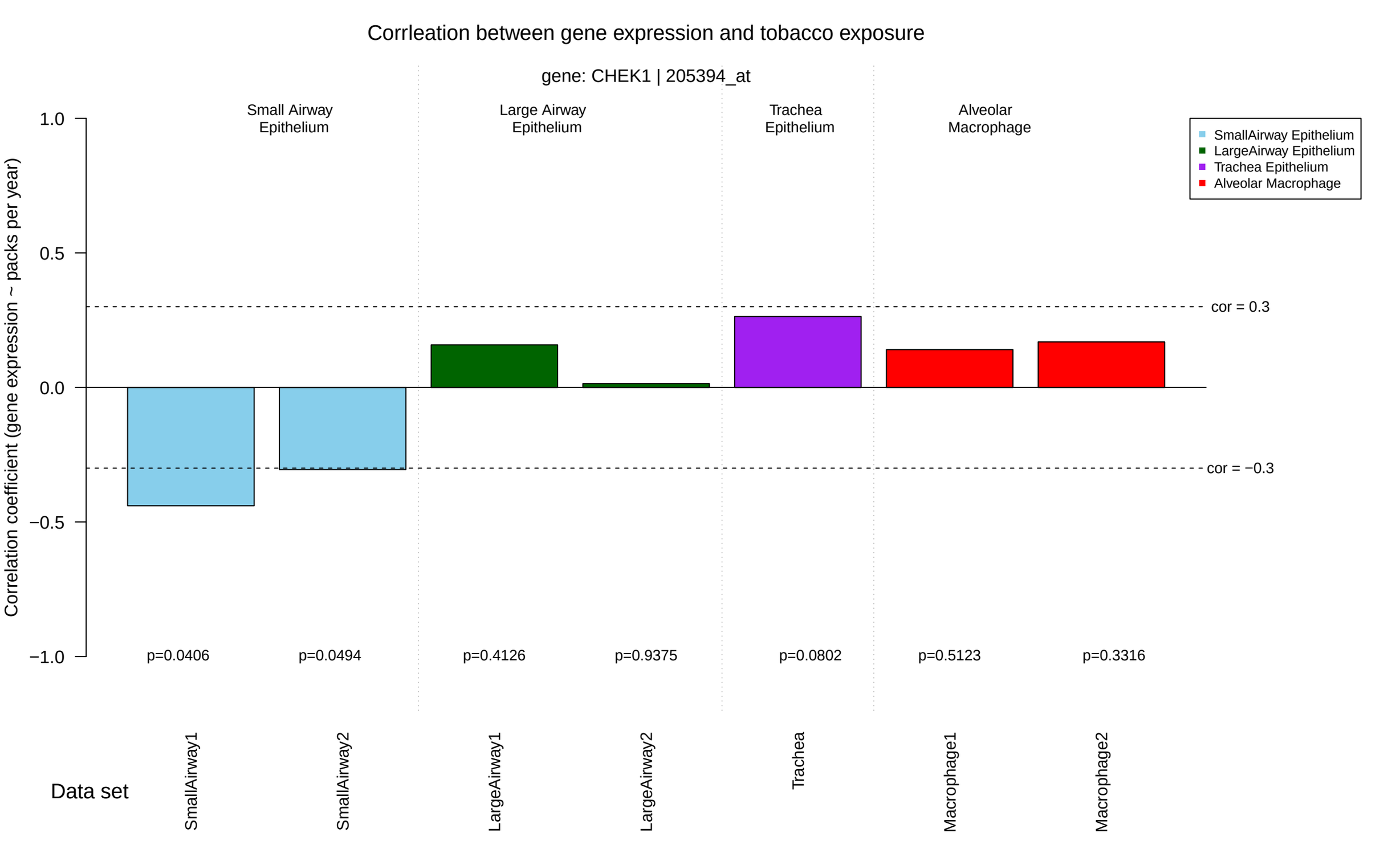

set: LargeAirway1 (p=0.4225)
CHEK1 | 205394_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8373)
CHEK1 | 205394_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3952)
CHEK1 | 205394_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9415)
CHEK1 | 205394_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3406)
CHEK1 | 205394_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CHEK1 | 205394_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0406
p=0.0494
p=0.4126
p=0.9375
p=0.0802
p=0.5123
p=0.3316
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2